

NETS 1500 Final Project User Manual
Modelling Prediction Markets through Graphs
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For a detailed project description and breakdown, please refer to the readme.txt (not the README.md file).

User Manual

1. Open the Java project folder in a code editor (i.e., Visual Studio Code)
2. Navigate into the project folder and compile the project by running this command:
javac \$(find src/main/java -name '*.java')
3. Run the program with no changes via the command
java -cp src/main/java edu.upenn.nets1500.kalshi.App
4. ****Note:** the program may take up to 30 seconds to run since there is a hefty underlying HTTP request being made to Kalshi's public API! The following is what you should see in the terminal after a few seconds:
 - a. The first demo is that of the graph traversals. Currently, it calls BFS and DFS on the first market fetched, but the user can change this in main in App.java by finding the line `printTraversalDemo(graph, markets.get(0).ticker());`
 - i. The current output is shown below, showing you the traversal order of markets with the ticker of the market in parentheses. Note that it often shows an almost identical market name but with a different ticker, showing that these are separate markets on Kalshi.

```
DEMO 1: Graph traversal demo start at Market:
Will Toronto be the 2026 AL East Division Winner
full title: Will Toronto be the 2026 AL East Division Winner
ticker: KXMLBALEAST-26-TOR

DEMO 1A: BFS order:
1. Will Toronto be the 2026 AL East Division Winner (KXMLBALEAST-26-TOR)
2. Will Tampa Bay be the 2026 AL East Division W... (KXMLBALEAST-26-TB)
3. Will New York Y be the 2026 AL East Division... (KXMLBALEAST-26-NYY)
DEMO 1B: DFS order:
1. Will Toronto be the 2026 AL East Division Winner (KXMLBALEAST-26-TOR)
2. Will Tampa Bay be the 2026 AL East Division W... (KXMLBALEAST-26-TB)
3. Will New York Y be the 2026 AL East Division... (KXMLBALEAST-26-NYY)
```

- b. The second demo is that of connected components. This requires no user input as it will always list all the connected components in the graph. However, the contents of the components may change as the Kalshi API fetches different markets every time the user runs the program. Output is shown below.

```

DEMO 2: Groups of connected components:
Component 1 (3 markets):
  Will Toronto be the 2026 AL East Division Winner
  full title: Will Toronto be the 2026 AL East Division Winner
  ticker: KONLBAL-EAST-26-TOR
  Will Tampa Bay be the 2026 AL East Division W...
  full title: Will Tampa Bay be the 2026 AL East Division Winner
  ticker: KONLBAL-EAST-26-TB
  Will New York Y be the 2026 AL East Division...
  full title: Will New York Y be the 2026 AL East Division Winner
  ticker: KONLBAL-EAST-26-NYY

Component 2 (2 markets):
  Will the United States Mint or the Department...
  full title: Will the United States Mint or the Department of the Treasury announce the official selection of a final design for a $1 coin featuring Donald Trump in 2026?
  ticker: KOTRUMPOOLLAR-26
  Will A Quiet Place Part III be released in the United States...
  full title: Will A Quiet Place Part III be released in the United States before Jan 1, 2028?
  ticker: KOMEDIARELEASEQUETPLACES-JAN28

Component 3 (3 markets):
  Will the price of natural gas get above $7.00 per million BTU...
  full title: Will the price of natural gas get above $7.00 per million BTU before January 1, 2027?
  ticker: KONGASMAX-26DEC31-P7.00
  Will the price of natural gas get above $6.00 per million BTU...
  full title: Will the price of natural gas get above $6.00 per million BTU before January 1, 2027?
  ticker: KONGASMAX-26DEC31-P6.00
  Will the price of natural gas get above $6.50 per million BTU...
  full title: Will the price of natural gas get above $6.50 per million BTU before January 1, 2027?
  ticker: KONGASMAX-26DEC31-P6.50

Component 4 (3 markets):
  All-leg parlay: Josh Hart: 6+; Tyrese Maxey: 6+; Stephen Castle: 10+...
  full title: yes Josh Hart: 6+,yes Tyrese Maxey: 6+,yes Stephen Castle: 10+,yes Karl-Anthony Towns: 15+,yes Mikal Bridges: 15+,yes OG Anunoby: 20+,yes Paul George: 20+,yes Tyrese Maxey: 30+,yes VJ Edgecombe: 15+
  ticker: KONVESPORTSMULTIGAMEEXTENDED-S20266FAEDCD08F8-166645FC9C3
  All-leg parlay: Josh Hart: 1+; VJ Edgecombe: 2+; Josh Hart: 6+ (+5 more)
  full title: yes Josh Hart: 1+,yes VJ Edgecombe: 2+,yes Josh Hart: 6+,yes Tyrese Maxey: 6+,yes Josh Hart: 8+,yes Jalen Brunson: 2+,yes Josh Hart: 3+,yes Tyrese Maxey: 2+
  ticker: KONVESPORTSMULTIGAMEEXTENDED-S2026A65F283341D-8311C82E8F9
  All-leg parlay: Anthony Edwards: 20+; Jaden McDaniels: 15+; Stephen C...
  full title: yes Anthony Edwards: 20+,yes Jaden McDaniels: 15+,yes Stephen Castle: 15+,yes Kelly Oubre Jr.: 15+,yes VJ Edgecombe: 15+
  ticker: KONVESPORTSMULTIGAMEEXTENDED-S2026BACE7B8114A-1EA451D39FA

Component 5 (3 markets):
  All-leg parlay: Colorado: A's; New York Y (+6 more)
  full title: yes Colorado, yes A's, yes New York Y, yes Cincinnati, yes Cleveland, yes Arizona, yes Texas, yes San Antonio, yes New York
  ticker: KONVECRSSCATEGORY-S20267CB01691E1B-A0BAEDDE16C
  All-leg parlay: New York M; Philadelphia; Baltimore (+2 more)
  full title: yes New York M, yes Philadelphia, yes Baltimore, yes Cleveland, yes Pittsburgh
  ticker: KONVESPORTSMULTIGAMEEXTENDED-S2026C391851DF8-26967FEDBC3
  All-leg parlay: New York M; Chicago C; Paul Skenes: 5+ (+1 more)
  full title: yes New York M, yes Chicago C, yes Paul Skenes: 5+, yes Minnesota
  ticker: KONVESPORTSMULTIGAMEEXTENDED-S202658003CE91E5-6310003E065

Component 6 (1 markets):
  All-leg parlay: Victor Wembanyama: 4+; Anthony Edwards: 25+; VJ Edgec...
  full title: yes Victor Wembanyama: 4+, yes Anthony Edwards: 25+, yes VJ Edgecombe: 15+, no San Antonio wins by over 19.5 points, yes Philadelphia wins by over 2.5 points, yes Anthony Edwards: 1+, no Over 230.5 points scored, yes Over 229.5 points scored
  ticker: KONVESPORTSMULTIGAMEEXTENDED-S202656AF8FB0421-95C6468BA63

```

c. The third demo is a ranking of the degrees for all the nodes in the graph. This will show all 15 markets and the number of degrees they have, in order of largest to smallest. Again, no user input is needed, and the output is included below.

```

DEMO 3: Degree ranking (in order of most connected to least connected Markets):
1.
  Will New York Y be the 2026 AL East Division...
  full title: Will New York Y be the 2026 AL East Division Winner
  ticker: KONLBAL-EAST-26-NYY
  degree: 2
2.
  Will Tampa Bay be the 2026 AL East Division M...
  full title: Will Tampa Bay be the 2026 AL East Division Winner
  ticker: KONLBAL-EAST-26-TB
  degree: 2
3.
  Will Toronto be the 2026 AL East Division Winner
  full title: Will Toronto be the 2026 AL East Division Winner
  ticker: KONLBAL-EAST-26-TOR
  degree: 2
4.
  All-leg parlay: Josh Hart: 6+; Tyrese Maxey: 6+; Stephen Castle: 10+...
  full title: yes Josh Hart: 6+,yes Tyrese Maxey: 6+,yes Stephen Castle: 10+,yes Karl-Anthony Towns: 15+,yes Mikal Bridges: 15+,yes OG Anunoby: 20+,yes Paul George: 20+,yes Tyrese Maxey: 30+,yes VJ Edgecombe: 15+
  ticker: KONVESPORTSMULTIGAMEEXTENDED-S20266FAEDCD08F8-166645FC9C3
  degree: 2
5.
  All-leg parlay: New York M; Philadelphia; Baltimore (+2 more)
  full title: yes New York M, yes Philadelphia, yes Baltimore, yes Cleveland, yes Pittsburgh
  ticker: KONVESPORTSMULTIGAMEEXTENDED-S2026C391851DF8-26967FEDBC3
  degree: 2
6.
  Will the price of natural gas get above $6.00 per million BTU...
  full title: Will the price of natural gas get above $6.00 per million BTU before January 1, 2027?
  ticker: KONGASMAX-26DEC31-P6.00
  degree: 2
7.
  Will the price of natural gas get above $6.50 per million BTU...
  full title: Will the price of natural gas get above $6.50 per million BTU before January 1, 2027?
  ticker: KONGASMAX-26DEC31-P6.50
  degree: 2
8.
  Will the price of natural gas get above $7.00 per million BTU...
  full title: Will the price of natural gas get above $7.00 per million BTU before January 1, 2027?
  ticker: KONGASMAX-26DEC31-P7.00
  degree: 2
9.
  Will A Quiet Place Part III be released in the United States...
  full title: Will A Quiet Place Part III be released in the United States before Jan 1, 2028?
  ticker: KOMEDIARELEASEQUETPLACES-JAN28
  degree: 1
10.
  All-leg parlay: Colorado: A's; New York Y (+6 more)
  full title: yes Colorado, yes A's, yes New York Y, yes Cincinnati, yes Cleveland, yes Arizona, yes Texas, yes San Antonio, yes New York
  ticker: KONVECRSSCATEGORY-S20267CB01691E1B-A0BAEDDE16C
  degree: 1
11.
  All-leg parlay: New York M; Chicago C; Paul Skenes: 5+ (+1 more)
  full title: yes New York M, yes Chicago C, yes Paul Skenes: 5+, yes Minnesota
  ticker: KONVESPORTSMULTIGAMEEXTENDED-S202658003CE91E5-6310003E065
  degree: 1
12.
  All-leg parlay: Josh Hart: 1+; VJ Edgecombe: 2+; Josh Hart: 6+ (+5 more)
  full title: yes Josh Hart: 1+,yes VJ Edgecombe: 2+,yes Josh Hart: 6+,yes Tyrese Maxey: 6+,yes Josh Hart: 8+,yes Jalen Brunson: 2+,yes Josh Hart: 3+,yes Tyrese Maxey: 2+
  ticker: KONVESPORTSMULTIGAMEEXTENDED-S2026A65F283341D-8311C82E8F9
  degree: 1
13.
  All-leg parlay: Anthony Edwards: 20+; Jaden McDaniels: 15+; Stephen C...
  full title: yes Anthony Edwards: 20+,yes Jaden McDaniels: 15+,yes Stephen Castle: 15+,yes Kelly Oubre Jr.: 15+,yes VJ Edgecombe: 15+
  ticker: KONVESPORTSMULTIGAMEEXTENDED-S2026BACE7B8114A-1EA451D39FA
  degree: 1
14.
  Will the United States Mint or the Department...
  full title: Will the United States Mint or the Department of the Treasury announce the official selection of a final design for a $1 coin featuring Donald Trump in 2026?
  ticker: KOTRUMPOOLLAR-26
  degree: 1
15.
  All-leg parlay: Victor Wembanyama: 4+; Anthony Edwards: 25+; VJ Edgec...
  full title: yes Victor Wembanyama: 4+, yes Anthony Edwards: 25+, yes VJ Edgecombe: 15+, no San Antonio wins by over 19.5 points, yes Philadelphia wins by over 2.5 points, yes Anthony Edwards: 1+, no Over 230.5 points scored, yes Over 229.5 points scored
  ticker: KONVESPORTSMULTIGAMEEXTENDED-S202656AF8FB0421-95C6468BA63
  degree: 0

```

d. The fourth demo is finding the shortest weighted path between two nodes. Currently, we have selected two nodes within the same connected component, ensuring a path exists. The user can change these inputs within the

printPathFindingDemo(MarketGraph graph, List<Market> component) in App.java by changing the values for startTicker and endTicker. The output as is shown below.

```
DEMO 4: Finding shortest weighted path:
Start market:
  Will Toronto be the 2026 AL East Division Winner
  full title: Will Toronto be the 2026 AL East Division Winner
  ticker: KXMLBALEAST-26-TOR
End market:
  Will New York Y be the 2026 AL East Division...
  full title: Will New York Y be the 2026 AL East Division Winner
  ticker: KXMLBALEAST-26-NYY

Shortest path:
1.
  Will Toronto be the 2026 AL East Division Winner
  full title: Will Toronto be the 2026 AL East Division Winner
  ticker: KXMLBALEAST-26-TOR
2.
  Will New York Y be the 2026 AL East Division...
  full title: Will New York Y be the 2026 AL East Division Winner
  ticker: KXMLBALEAST-26-NYY
```

- e. The fifth demo finds the top neighbors for each market by calling neighborsOf. The user can see all of the neighbors for all fifteen market and their similarity score. No user input required, and the output is shown below.

```
DEMO 5: Top neighbors for each market:

Will Toronto be the 2026 AL East Division Winner
full title: Will Toronto be the 2026 AL East Division Winner
ticker: KXMLBALEAST-26-TOR
1.
  Will Tampa Bay be the 2026 AL East Division W...
  full title: Will Tampa Bay be the 2026 AL East Division Winner
  ticker: KXMLBALEAST-26-TB
  score: 0.88
2.
  Will New York Y be the 2026 AL East Division...
  full title: Will New York Y be the 2026 AL East Division Winner
  ticker: KXMLBALEAST-26-NYY
  score: 0.75

Will Tampa Bay be the 2026 AL East Division W...
full title: Will Tampa Bay be the 2026 AL East Division Winner
ticker: KXMLBALEAST-26-TB
1.
  Will Toronto be the 2026 AL East Division Winner
  full title: Will Toronto be the 2026 AL East Division Winner
  ticker: KXMLBALEAST-26-TOR
  score: 0.88
2.
  Will New York Y be the 2026 AL East Division...
  full title: Will New York Y be the 2026 AL East Division Winner
  ticker: KXMLBALEAST-26-NYY
  score: 0.71

Will New York Y be the 2026 AL East Division...
full title: Will New York Y be the 2026 AL East Division Winner
ticker: KXMLBALEAST-26-NYY
1.
  Will Toronto be the 2026 AL East Division Winner
  full title: Will Toronto be the 2026 AL East Division Winner
  ticker: KXMLBALEAST-26-TOR
  score: 0.75
2.
  Will Tampa Bay be the 2026 AL East Division W...
  full title: Will Tampa Bay be the 2026 AL East Division Winner
  ticker: KXMLBALEAST-26-TB
  score: 0.71

Will the United States Mint or the Department...
full title: Will the United States Mint or the Department of the Treasury announce the official selection of a final design for a $1 coin featuring Donald Trump in 2027?
ticker: KXTRUMPOLLAR-26
1.
  Will A Quiet Place Part III be released in the United States...
  full title: Will A Quiet Place Part III be released in the United States before Jan 1, 2027?
  ticker: KQWIDIARELEASEQUIETPLACES-26J03
  score: 0.31

Will the price of natural gas get above $7.00 per million BTU...
full title: Will the price of natural gas get above $7.00 per million BTU before January 1, 2027?
ticker: KONGASMAX-26DEC31-97.00
1.
  Will the price of natural gas get above $6.00 per million BTU...
  full title: Will the price of natural gas get above $6.00 per million BTU before January 1, 2027?
  ticker: KONGASMAX-26DEC31-P6.00
  score: 0.94
2.
  Will the price of natural gas get above $6.50 per million BTU...
  full title: Will the price of natural gas get above $6.50 per million BTU before January 1, 2027?
  ticker: KONGASMAX-26DEC31-P6.50
  score: 0.83

Will the price of natural gas get above $6.50 per million BTU...
full title: Will the price of natural gas get above $6.50 per million BTU before January 1, 2027?
ticker: KONGASMAX-26DEC31-P6.50
1.
  Will the price of natural gas get above $6.00 per million BTU...
  full title: Will the price of natural gas get above $6.00 per million BTU before January 1, 2027?
  ticker: KONGASMAX-26DEC31-P6.00
  score: 0.89
```